

Results

Aug 12, 2024

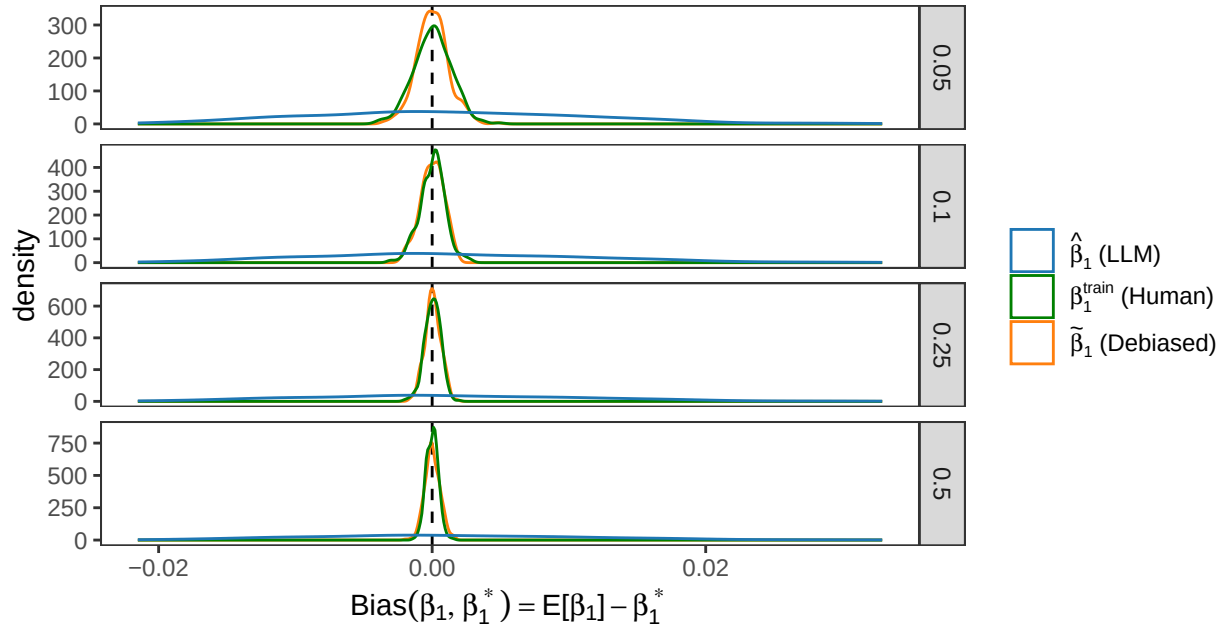
Proxy on the LHS

Number of combinations = 1370/1440

Bootstrap = Bayesian

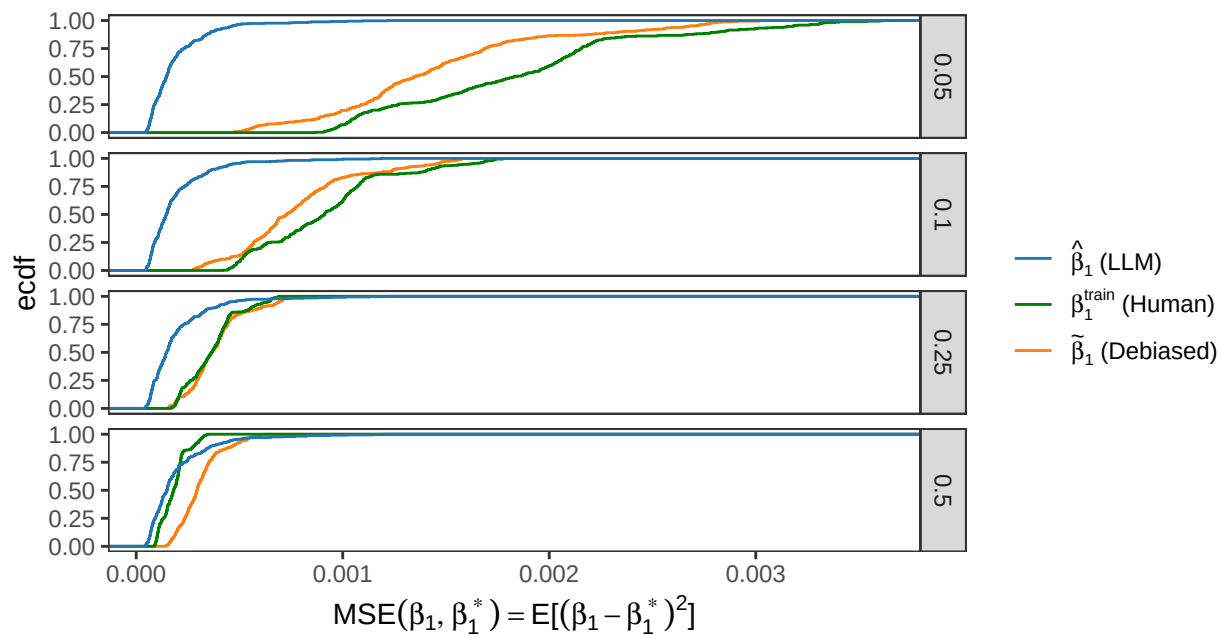
Density Plots of Bias(β_1, β_1^*) by Training Proportion

LHS Regression: $1\{Y = y\} = \beta_0 + \beta_1 V$



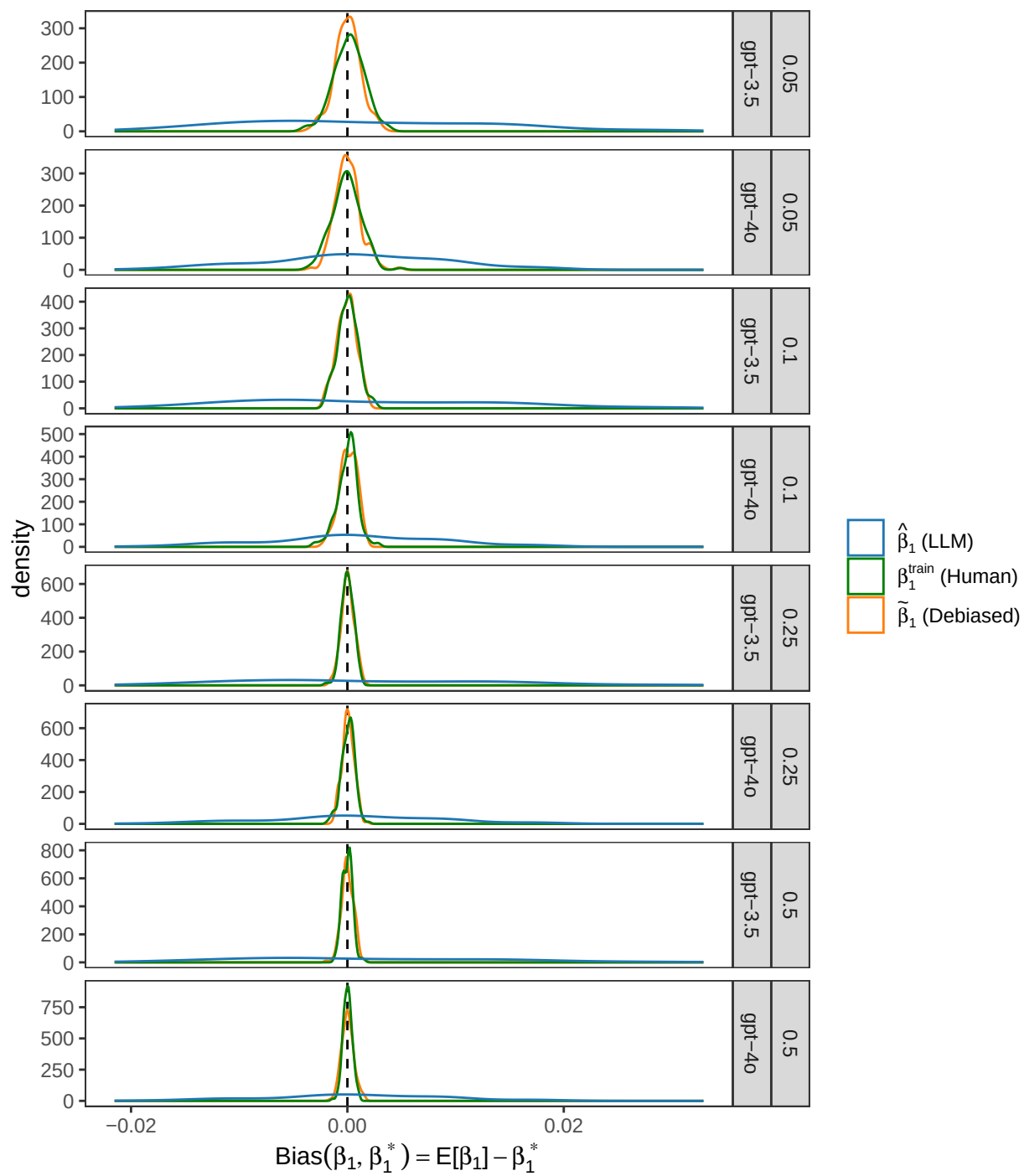
Empirical CDF of $\text{MSE}(\beta_1, \beta_1^*)$ by Training Proportion

LHS Regression: $1\{Y = y\} = \beta_0 + \beta_1 V$



Density Plots of Bias(β_1, β_1^*) by Model & Training Proportion

LHS Regression: $1\{Y = y\} = \beta_0 + \beta_1 V$



Empirical CDF of $\text{MSE}(\beta_1, \beta_1^*)$ by Model & Training Proportion

LHS Regression: $1\{Y = y\} = \beta_0 + \beta_1 V$

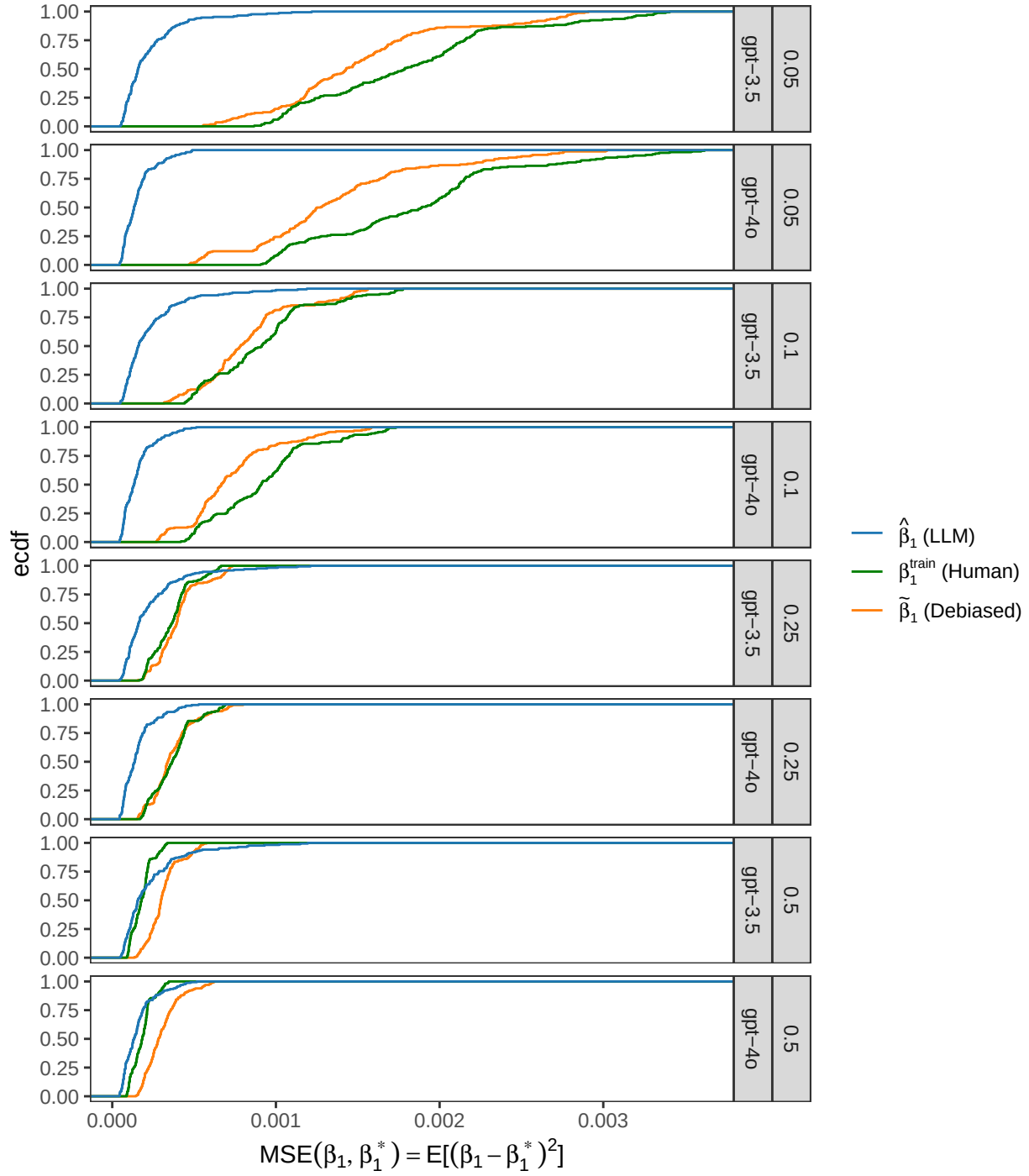


Table 1: Bias Summary Statistics by Training Proportion

Train Proportion	Regression	Mean	SD	Median	5%	95%
0.05	Yllm_V	0.00125	0.01027	0.00055	-0.01465	0.01816
0.05	test_Ytilde_V	0.00003	0.00119	-0.00002	-0.00195	0.00218
0.05	train_Yhuman_V	0.00004	0.00139	0.00006	-0.00223	0.00221
0.10	Yllm_V	0.00135	0.01030	0.00045	-0.01416	0.01850
0.10	test_Ytilde_V	0.00000	0.00086	0.00002	-0.00153	0.00134
0.10	train_Yhuman_V	0.00002	0.00095	0.00013	-0.00159	0.00141
0.25	Yllm_V	0.00123	0.01032	0.00034	-0.01459	0.01828
0.25	test_Ytilde_V	0.00005	0.00059	0.00004	-0.00092	0.00102
0.25	train_Yhuman_V	0.00004	0.00059	0.00007	-0.00091	0.00095
0.50	Yllm_V	0.00125	0.01036	0.00050	-0.01450	0.01850
0.50	test_Ytilde_V	0.00002	0.00056	-0.00002	-0.00086	0.00094
0.50	train_Yhuman_V	0.00000	0.00044	0.00003	-0.00068	0.00070

Table 2: MSE Summary Statistics by Training Proportion

Train Proportion	Regression	Mean	SD	Median	5%	95%
0.05	Yllm_V	0.00019	0.00016	0.00015	0.00006	0.00046
0.05	test_Ytilde_V	0.00144	0.00057	0.00135	0.00058	0.00268
0.05	train_Yhuman_V	0.00184	0.00064	0.00183	0.00098	0.00318
0.10	Yllm_V	0.00019	0.00016	0.00015	0.00006	0.00047
0.10	test_Ytilde_V	0.00078	0.00030	0.00073	0.00033	0.00145
0.10	train_Yhuman_V	0.00092	0.00032	0.00091	0.00048	0.00161
0.25	Yllm_V	0.00019	0.00016	0.00015	0.00006	0.00045
0.25	test_Ytilde_V	0.00039	0.00014	0.00037	0.00019	0.00069
0.25	train_Yhuman_V	0.00037	0.00013	0.00037	0.00019	0.00063
0.50	Yllm_V	0.00019	0.00016	0.00015	0.00006	0.00047
0.50	test_Ytilde_V	0.00031	0.00010	0.00029	0.00017	0.00053
0.50	train_Yhuman_V	0.00018	0.00006	0.00018	0.00010	0.00031

Table 3: Coverage Summary Statistics by Training Proportion

Train Proportion	Regression	Mean	SD	Median	5%	95%
0.05	Yllm_V	0.80793	0.16441	0.884	0.4615	0.95100
0.05	test_Ytilde_V	0.92851	0.01128	0.929	0.9100	0.94600
0.05	train_Yhuman_V	0.94505	0.00781	0.945	0.9329	0.95800
0.10	Yllm_V	0.80858	0.16428	0.882	0.4596	0.95000
0.10	test_Ytilde_V	0.94084	0.00773	0.941	0.9279	0.95300
0.10	train_Yhuman_V	0.94836	0.00713	0.949	0.9360	0.95910
0.25	Yllm_V	0.80728	0.16370	0.885	0.4470	0.95015
0.25	test_Ytilde_V	0.94586	0.00723	0.946	0.9340	0.95715
0.25	train_Yhuman_V	0.94920	0.00725	0.949	0.9370	0.96100
0.50	Yllm_V	0.80583	0.16686	0.883	0.4440	0.95100
0.50	test_Ytilde_V	0.94686	0.00708	0.947	0.9340	0.95800
0.50	train_Yhuman_V	0.95009	0.00670	0.950	0.9390	0.96100

Table 4: Bias Summary Statistics by Model and Training Proportion

Train Proportion	Model	Regression	Mean	SD	Median	5%	95%
0.05	gpt-3.5	Yllm_V	0.00177	0.01189	0.00043	-0.01613	0.02127
0.05	gpt-3.5	test_Ytilde_V	0.00002	0.00122	0.00007	-0.00207	0.00218
0.05	gpt-3.5	train_Yhuman_V	0.00015	0.00143	0.00020	-0.00218	0.00240
0.05	gpt-4o	Yllm_V	0.00072	0.00831	0.00082	-0.01352	0.01512
0.05	gpt-4o	test_Ytilde_V	0.00003	0.00117	-0.00005	-0.00166	0.00217
0.05	gpt-4o	train_Yhuman_V	-0.00008	0.00134	-0.00011	-0.00223	0.00212
0.10	gpt-3.5	Yllm_V	0.00199	0.01202	0.00014	-0.01533	0.02329
0.10	gpt-3.5	test_Ytilde_V	-0.00007	0.00089	0.00000	-0.00171	0.00140
0.10	gpt-3.5	train_Yhuman_V	0.00005	0.00093	0.00010	-0.00159	0.00147
0.10	gpt-4o	Yllm_V	0.00069	0.00814	0.00084	-0.01314	0.01522
0.10	gpt-4o	test_Ytilde_V	0.00008	0.00082	0.00007	-0.00141	0.00129
0.10	gpt-4o	train_Yhuman_V	0.00000	0.00096	0.00013	-0.00160	0.00140
0.25	gpt-3.5	Yllm_V	0.00182	0.01196	-0.00004	-0.01605	0.02175
0.25	gpt-3.5	test_Ytilde_V	0.00004	0.00061	0.00002	-0.00095	0.00104
0.25	gpt-3.5	train_Yhuman_V	0.00001	0.00057	-0.00001	-0.00083	0.00091
0.25	gpt-4o	Yllm_V	0.00063	0.00829	0.00047	-0.01352	0.01592
0.25	gpt-4o	test_Ytilde_V	0.00006	0.00056	0.00005	-0.00091	0.00100
0.25	gpt-4o	train_Yhuman_V	0.00007	0.00062	0.00012	-0.00112	0.00096
0.50	gpt-3.5	Yllm_V	0.00178	0.01208	0.00016	-0.01617	0.02303
0.50	gpt-3.5	test_Ytilde_V	0.00002	0.00056	-0.00003	-0.00088	0.00086
0.50	gpt-3.5	train_Yhuman_V	-0.00001	0.00047	0.00008	-0.00076	0.00069
0.50	gpt-4o	Yllm_V	0.00071	0.00823	0.00060	-0.01337	0.01580
0.50	gpt-4o	test_Ytilde_V	0.00002	0.00057	0.00000	-0.00084	0.00100
0.50	gpt-4o	train_Yhuman_V	0.00002	0.00042	0.00001	-0.00063	0.00070

Table 5: MSE Summary Statistics by Model and Training Proportion

Train Proportion	Model	Regression	Mean	SD	Median	5%	95%
0.05	gpt-3.5	Yllm_V	0.00023	0.00020	0.00016	0.00007	0.00058
0.05	gpt-3.5	test_Ytilde_V	0.00151	0.00056	0.00143	0.00071	0.00273
0.05	gpt-3.5	train_Yhuman_V	0.00182	0.00063	0.00179	0.00098	0.00316
0.05	gpt-4o	Yllm_V	0.00016	0.00010	0.00013	0.00006	0.00038
0.05	gpt-4o	test_Ytilde_V	0.00137	0.00056	0.00126	0.00054	0.00254
0.05	gpt-4o	train_Yhuman_V	0.00186	0.00065	0.00189	0.00097	0.00317
0.10	gpt-3.5	Yllm_V	0.00023	0.00020	0.00016	0.00007	0.00066
0.10	gpt-3.5	test_Ytilde_V	0.00083	0.00030	0.00078	0.00039	0.00147
0.10	gpt-3.5	train_Yhuman_V	0.00092	0.00033	0.00091	0.00048	0.00159
0.10	gpt-4o	Yllm_V	0.00015	0.00010	0.00013	0.00006	0.00037
0.10	gpt-4o	test_Ytilde_V	0.00072	0.00030	0.00067	0.00030	0.00131
0.10	gpt-4o	train_Yhuman_V	0.00092	0.00032	0.00091	0.00048	0.00160
0.25	gpt-3.5	Yllm_V	0.00023	0.00020	0.00016	0.00007	0.00059
0.25	gpt-3.5	test_Ytilde_V	0.00040	0.00014	0.00039	0.00020	0.00069
0.25	gpt-3.5	train_Yhuman_V	0.00036	0.00012	0.00036	0.00019	0.00061
0.25	gpt-4o	Yllm_V	0.00016	0.00010	0.00014	0.00006	0.00039
0.25	gpt-4o	test_Ytilde_V	0.00037	0.00014	0.00034	0.00018	0.00068
0.25	gpt-4o	train_Yhuman_V	0.00037	0.00013	0.00037	0.00019	0.00065
0.50	gpt-3.5	Yllm_V	0.00023	0.00020	0.00016	0.00007	0.00065
0.50	gpt-3.5	test_Ytilde_V	0.00031	0.00010	0.00030	0.00017	0.00052
0.50	gpt-3.5	train_Yhuman_V	0.00018	0.00006	0.00018	0.00010	0.00031
0.50	gpt-4o	Yllm_V	0.00016	0.00010	0.00013	0.00006	0.00039
0.50	gpt-4o	test_Ytilde_V	0.00030	0.00011	0.00028	0.00017	0.00055
0.50	gpt-4o	train_Yhuman_V	0.00018	0.00006	0.00018	0.00010	0.00031

Table 6: Coverage Summary Statistics by Model and Training Proportion

Train Proportion	Model	Regression	Mean	SD	Median	5%	95%
0.05	gpt-3.5	Yllm_V	0.76126	0.18953	0.8180	0.37250	0.94900
0.05	gpt-3.5	test_Ytilde_V	0.93018	0.01110	0.9310	0.91250	0.94750
0.05	gpt-3.5	train_Yhuman_V	0.94537	0.00808	0.9450	0.93300	0.95800
0.05	gpt-4o	Yllm_V	0.85543	0.11680	0.9130	0.63500	0.95265
0.05	gpt-4o	test_Ytilde_V	0.92682	0.01125	0.9280	0.90900	0.94400
0.05	gpt-4o	train_Yhuman_V	0.94473	0.00754	0.9450	0.93300	0.95665
0.10	gpt-3.5	Yllm_V	0.75888	0.18775	0.8055	0.37585	0.94345
0.10	gpt-3.5	test_Ytilde_V	0.94110	0.00744	0.9410	0.92855	0.95300
0.10	gpt-3.5	train_Yhuman_V	0.94771	0.00707	0.9470	0.93700	0.95845
0.10	gpt-4o	Yllm_V	0.85978	0.11593	0.9160	0.64720	0.95400
0.10	gpt-4o	test_Ytilde_V	0.94057	0.00803	0.9410	0.92730	0.95300
0.10	gpt-4o	train_Yhuman_V	0.94903	0.00715	0.9500	0.93600	0.96000
0.25	gpt-3.5	Yllm_V	0.75974	0.18717	0.7965	0.37555	0.94790
0.25	gpt-3.5	test_Ytilde_V	0.94543	0.00703	0.9450	0.93455	0.95745
0.25	gpt-3.5	train_Yhuman_V	0.94933	0.00724	0.9490	0.93700	0.96100
0.25	gpt-4o	Yllm_V	0.85653	0.11674	0.9135	0.64400	0.95075
0.25	gpt-4o	test_Ytilde_V	0.94631	0.00742	0.9470	0.93400	0.95700
0.25	gpt-4o	train_Yhuman_V	0.94907	0.00727	0.9490	0.93800	0.96000
0.50	gpt-3.5	Yllm_V	0.75701	0.19183	0.8230	0.37600	0.94750
0.50	gpt-3.5	test_Ytilde_V	0.94733	0.00712	0.9480	0.93350	0.95850
0.50	gpt-3.5	train_Yhuman_V	0.95002	0.00653	0.9500	0.93950	0.96000
0.50	gpt-4o	Yllm_V	0.85613	0.11733	0.9125	0.62300	0.95200
0.50	gpt-4o	test_Ytilde_V	0.94638	0.00703	0.9460	0.93400	0.95700
0.50	gpt-4o	train_Yhuman_V	0.95017	0.00689	0.9505	0.93825	0.96175